PMI, PPMI, and SVD

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The Core Problem

Raw Co-occurrence Counts

- Frequency $\neq$ Association
- "the" co-occurs with everything
- Common words dominate
- Rare pairs underrepresented

What We Want

- Meaningful association strength
- Surprise: observed vs. expected
- Scale-independent measure
- Semantic relatedness signal

Question: How do we transform raw counts into semantic similarity?

The Intuition Behind PMI

Core Idea

PMI measures how much **more** (or less) often two words co-occur than we would expect **by chance**.

High PMI

"cherry" + "pie"
Occur together far more than expected

Zero PMI

"the" + "dog"
Occur exactly as often as expected

Negative PMI

"cherry" + "democracy"
Occur less than expected

Pointwise Mutual Information (PMI)

Definition

$$\text{PMI}(w, c) = \log_2 \frac{P(w, c)}{P(w) \cdot P(c)}$$

Where:

$P(w, c) =$ joint probability of word w and context c co-occurring

$P(w) =$ marginal probability of word w

$P(c) =$ marginal probability of context c

Interpretation

$\text{PMI} > 0$: w and c co-occur **more** than chance

$\text{PMI} < 0$: w and c co-occur **less** than chance

$\text{PMI} = 0$: w and c co-occur exactly as expected

Computing PMI from Counts

Starting with a co-occurrence matrix with N total observations:

Probability Estimates

$$P(w, c) = \frac{\#(w, c)}{N}$$

$$P(w) = \frac{\#(w)}{N} \quad \text{where } \#(w) = \sum_{c} \#(w, c)$$

$$P(c) = \frac{\#(c)}{N} \quad \text{where } \#(c) = \sum_{w} \#(w, c)$$

PMI in Terms of Counts

$$\text{PMI}(w, c) = \log_2 \frac{\#(w, c) \cdot N}{\#(w) \cdot \#(c)}$$

The Problems with PMI

Zero Counts $\rightarrow -\infty$

When $\#(w, c) = 0$:

$$\log_2(0) = -\infty$$

Undefined values break downstream computations.

Rare Word Bias

PMI inflates scores for rare words that happen to co-occur.

A single co-occurrence of rare words yields extremely high PMI.

Negative Values Are Problematic

Negative PMI indicates "less than chance" co-occurrence, but this signal is noisy and unreliable. Most word pairs have negative PMI simply because they're unrelated—not informatively anti-correlated.

Positive PMI (PPMI)

Definition

$$\text{PPMI}(w, c) = \max\big(\text{PMI}(w, c),\ 0\big)$$

A simple but effective fix:

- Replace all negative PMI values with zero
- Handles the $-\infty$ problem for zero counts
- Focuses on positive associations (informationally richer)
- Creates sparse matrices (most entries are 0)

Standard Practice

PPMI matrices are the standard input for SVD-based dimensionality reduction.

PPMI Example

Raw Co-occurrence Counts

	computer	data	the
information	5	8	20
digital	12	3	15
ancient	1	1	25

$\Rightarrow$

PPMI Matrix

	computer	data	the
information	0.32	1.41	0
digital	1.58	0.12	0
ancient	0	0	0.23

What the PPMI tells us:

- "digital" and "computer" have strong association (1.58)
- "information" and "data" are meaningfully related (1.41)
- Zeros indicate expected or below-expected co-occurrence

The Dimensionality Problem

Scale

PPMI matrices are huge: $|V| \times |V|$ where $|V|$ can be 50,000+ words

Storage

$50\text{K} \times 50\text{K}$
$= 2.5 \times 10^9$ cells

Even sparse: massive

Noise

Most dimensions capture noise, not signal

Sparse data = unreliable estimates

Computation

Similarity calculations expensive

$O(|V|)$ per comparison

Solution

Reduce to 100–500 dense dimensions that capture the essential structure.

Singular Value Decomposition

The mathematical backbone of dimensionality reduction

SVD: The Factorization

Singular Value Decomposition

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

Where $\mathbf{M}$ is our $m \times n$ PPMI matrix:

$\mathbf{U}$ $(m \times m)$	$\mathbf{\Sigma}$ $(m \times n)$	$\mathbf{V}^\top$ $(n \times n)$
Left singular vectors	Diagonal matrix	Right singular vectors
Word representations	**Singular values**	**Context representations**
(rows = words)	(importance weights)	(columns = contexts)

Singular values in $\mathbf{\Sigma}$ are ordered: $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \cdots \geq 0$

Truncated SVD: The Key Insight

Key Idea

Keep only the top k singular values and their vectors $\rightarrow$ Best rank-k approximation

Truncated Form

$$\mathbf{M}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top \approx \mathbf{M}$$

Benefits of truncation:

- **Dimensionality reduction:** $|V|$ dimensions $\rightarrow$ k dimensions (typically 100–500)
- **Noise removal:** Small singular values often capture noise
- **Generalization:** Smooths over sparse observations
- **Latent structure:** Reveals hidden semantic dimensions

Why Does SVD Work for Semantics?

The Linguistic Insight

Words that never directly co-occur can still be similar.

"boat" and "ship" may share contexts like "sail", "ocean", "harbor"

SVD discovers these latent relationships by factoring the matrix into shared dimensions.

The Mathematical Property

Eckart–Young Theorem:

$\mathbf{M}_k$ is the best rank-k approximation of $\mathbf{M}$ under Frobenius norm:

$$\min_{\text{rank}(\mathbf{X})=k} \|\mathbf{M} - \mathbf{X}\|_F$$

Optimally preserves the most variance in the data.

Extracting Word Vectors

After truncated SVD, we have several options for word vectors:

Option 1: $\mathbf{U}_k$
Use left singular vectors directly. Each row is a word's k-dimensional vector.

Option 2: $\mathbf{U}_k\mathbf{\Sigma}_k$
Weight by singular values. Dimensions with more variance get more weight.

Option 3: $\mathbf{U}_k\mathbf{\Sigma}_k^p$
Power weighting (p often 0.5). Balances raw vectors and full weighting.

Common Practice

$\mathbf{U}_k\mathbf{\Sigma}_k^{0.5}$ (square root weighting) often performs best empirically.

LSA and LSI: The Precursors

Latent Semantic Analysis

Deerwester et al., 1990

Applied SVD to term-document matrices for information retrieval.

Key insight: Reduce dimensions to capture latent semantic structure.

Latent Semantic Indexing

Same technique, different name.

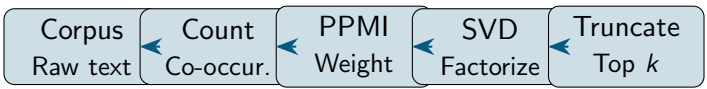
Used for document retrieval and similarity.

Documents and queries projected into same low-dimensional space.

Modern DS Word Vectors

Same math, different matrix. Instead of term-document, we use word-context (PPMI) matrices. The core idea—that SVD reveals latent semantic dimensions—remains the same.

The Complete Pipeline



Result: Dense Word Vectors (100–500 dimensions)

Each word represented as a point in semantic space where:

$$\cos(\mathbf{v}_{\text{word}_1}, \mathbf{v}_{\text{word}_2}) \approx \text{semantic similarity}$$

Practical Considerations

Context Window

Symmetric windows of 2–10 words.
Smaller = syntactic
Larger = topical

Dimensions (k)

Typically 100–500.
Diminishing returns beyond 500.

Vocabulary Size

Top 50K–200K words.
Filter rare words ($<$ 5–100 occurrences).

Weighting

Distance weighting: closer words count more ($1/d$ scaling).

Evaluating Word Vectors

Intrinsic Evaluation

Word Similarity

SimLex-999, WordSim-353

Correlation with human judgments

Analogy

"king − man + woman = queen"

Google analogy dataset

Extrinsic Evaluation

Downstream Tasks

Named Entity Recognition

Sentiment Analysis

Parsing

Use as Features

Initialize neural models

Concatenate with other features

Caveat

Intrinsic performance doesn't always predict extrinsic performance!

Looking Ahead: Limitations

What SVD-based methods can't do:

- **Composition:** How do we combine word vectors into phrase/sentence meaning?
- **Structure:** Word order and syntax are lost in the bag-of-words co-occurrence
- **Polysemy:** One vector per word, regardless of sense
- **Scalability:** Full SVD is $O(mn^2)$—expensive for large vocabularies

Coming Next

Vector composition models $\rightarrow$ Holographic Reduced Representations (HRR) for structure-preserving binding

Summary

- **PMI** transforms raw counts into association measures:

$$\text{PMI}(w, c) = \log_2 \frac{P(w, c)}{P(w) \cdot P(c)}$$

- **PPMI** handles negative/undefined values: $\text{PPMI} = \max(\text{PMI}, 0)$
- **SVD** provides optimal low-rank approximation: $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$
- **Truncated SVD** reduces noise and reveals latent structure
- **Result:** Dense word vectors capturing distributional semantics